Imperial College London

MEng (Engineering) Examination 2020

Year 1

77/100

AERO40006 Introduction to Mathematics

Thursday 15th October 2020: 11.00 to 12.00
[1 hour examination + max 30 minutes upload time]

OPEN BOOK EXAMINATION

There are FIVE questions. All questions carry equal weight.
Candidates may obtain full marks for complete answers to ***ALL*** questions.

A datasheet is provided.

This is an open book assessment. We have worked hard to create exams that assesses synthesis of knowledge rather than factual recall. Thus, access to the internet, notes or other sources of factual information in the time provided might not be helpful and may well limit your time to successfully synthesise the answers required.

Where individual questions rely more on factual recall and may therefore be less discriminatory in an open book context, we may compare the performance on these questions to similar style questions in previous years and we may scale or ignore the marks associated with such questions or parts of the questions.

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Section A

Question 1

Part (a)

Calculate the derivative of $y(x) = 5^x$. 50 [50%]

Part (b)

Find dy/dx when $y(x)$ is implicitely given by $y = x^2 + \sin(xy)$. 50 [50%]

(a) $\ln[y(x)] = \ln 5^x = x \ln 5$ $\quad \therefore \frac{1}{y}\frac{6y}{6x} = \ln 5$

$\therefore \frac{6y}{6x} = y \ln 5 = 5^x \ln 5$

(b) $\frac{dy}{dx} = 2x + y\cos(xy) + x\frac{dy}{dx}\cos(xy)$

$\therefore (1 - x\cos(xy))\frac{dy}{dx} = 2x + y\cos(xy)$

$\therefore \frac{dy}{dx} = \frac{2x + y\cos(xy)}{1 - x\cos(xy)}$

Question 2

Part (a)

Evaluate the integral $\int x \ln(x) dx$. 50 [50%]

Part (b)

The Fourier coefficient, a_n, for a triangular waveform is given by

$$a_n = \frac{1}{\pi^2 n^2} \int_0^{2\pi} (\omega t) \cos(n\omega t) d(\omega t)$$

where n is a positive integer. Find a_n. [50%]

(a) For $\int \ln(x) dx$, $\int \ln(x) dx = x\ln(x) - x + C$

	D	I
+	x	$\ln(x)$
−	1	$x\ln(x) - x$
+	0	...

$\therefore \int x\ln(x)dx = x^2\ln(x) - x^2 - \int x\ln(x) - x \, dx + C$

$= x^2\ln(x) - x^2 - \int x\ln(x)dx + \frac{1}{2}x^2 + C$

$\therefore 2\int x\ln(x)dx = x^2\ln(x) - \frac{1}{2}x^2 + C$

$\therefore \int x\ln(x)dx = \frac{1}{2}x^2\ln(x) - \frac{1}{4}x^2 + C$

Question 2

Part (a)

Evaluate the integral $\int x \ln(x) dx$. [50%]

Part (b)

The Fourier coefficient, a_n, for a triangular waveform is given by

$$a_n = \frac{1}{\pi^2 n^2} \int_0^{2\pi} (\omega t) \cos(n\omega t) d(\omega t)$$

where n is a positive integer. Find a_n. 45 [50%]

(b) $a_n = \frac{1}{\pi^2 n^2} \int_0^{2\pi} (\omega t) \cos(n\omega t)\, d(\omega t)$

Let $\omega t = x$

$\therefore \int_0^{2\pi} (\omega t) \cos(n\omega t)\, d(\omega t) = \int_0^{2\pi} x \cos(nx)\, dx$

	D	I
+	x	$\cos(nx)$
−	1	$\frac{1}{n}\sin(nx)$
+	0	$-\frac{1}{n^2}\cos(nx)$

$= \left[\frac{x}{n}\sin(nx) + \frac{1}{n^2}\cos(nx) \right]_0^{2\pi}$

$= \frac{2\pi}{n}\sin(2n\pi) + \frac{1}{n^2}\cos(2n\pi) \cancel{+} \frac{1}{n^2}$ $-\frac{1}{n^2}$

$= \frac{2}{n^2}$ $= 0$

$\therefore a_n = \frac{1}{\pi^2 n^2} \frac{2}{n^2} = \frac{2}{\pi^2 n^4}$ 0

Question 3

Express $I_n = \int (x^2+3)^n dx$ as a function of $I_{n-1},\ n$ and x. 0 [100%]

$$I_{n-1} = \int (x^2+3)^{n-1}\,dx = \frac{1}{2nx}(x^2+3)^n + C$$

$$I_n = \int (x^2+3)^n\,dx = \frac{1}{2(n+1)x}(x^2+3)^{n+1} + C$$

$$\therefore \frac{I_n}{I_{n-1}} = \frac{(2nx)(x^2+3)^{n+1}}{2(n+1)x(x^2+3)^n} = \frac{n}{n+1}(x^2+3)$$

$$\therefore I_n = \frac{n}{n+1}(x^2+3)\,I_{n-1}$$

$$I_n = x(x^2+3)^n - \int 2nx^2(x^2+3)^{n-1}\,dx$$

	D	I
+	$(x^2+3)^n$	1
−	$2nx(x^2+3)^{n-1}$	x

$$= x(x^2+3)^n - 2n\int (x^2+3-3)(x^2+3)^{n-1}\,dx$$

$$= x(x^2+3)^n - 2n\int (x^2+3)^n\,dx + 6n\int (x^2+3)^{n-1}\,dx$$

$$= x(x^2+3)^n - 2nI_n + 6nI_{n-1}$$

$$\therefore (2n+1)I_n = x(x^2+3)^n + 6nI_{n-1}$$

$$\therefore I_n = \frac{x(x^2+3)^n}{2n+1} + \frac{6n}{2n+1}I_{n-1}$$

Section B

Question 4

Part (a)

Split the expression $\frac{x+5}{x^2-2x+1}$ into partial fractions. 50 [50%]

Part (b)

For which x the series $\sum_{n\geq 1} \frac{(x-2)^n}{n}$ converge? 40 [50%]

(a) $\frac{x+5}{x^2-2x+1} = \frac{A}{(x-1)^2} + \frac{B}{x-1}$

$\therefore A + B(x-1) = x+5$

$\therefore Bx + (A-B) = x+5 \qquad \therefore B=1, A=6$

$\therefore \frac{x+5}{x^2+2x+1} = \frac{6}{(x-1)^2} + \frac{1}{x-1}$

(b) $\lim_{n\to\infty} \frac{\left|\frac{(x-2)^{n+1}}{n+1}\right|}{\left|\frac{(x-2)^n}{n}\right|} = \lim_{n\to\infty} \left| \frac{n}{n+1}(x-2) \right| = |x-2|$

If $|x-2| < 1$, series converge

$-1 < x-2 < 1$

$\therefore 1 < x < 3$

...

If $|x-2| = 1$, $x = \pm 3$

$\therefore$ for $x=3$, $\sum_{n\geq 1} \frac{(x-2)^n}{n} = \sum_{n\geq 1} \frac{1}{n} \longrightarrow$ diverge

for $x=1$, $\sum_{n\geq 1} \frac{(x-2)^n}{n} = \sum_{n\geq 1} \frac{(-1)^n}{n} \longrightarrow$ converge

$\therefore 1 \leq x < 3$

Question 5

Part (a)

Evaluate the limit $\lim_{x \to 0} \frac{e^{\sin(x)} - 1 - x}{x^2}$. 50 [50%]

Part (b)

Evaluate the limit $\lim_{x \to \infty} \frac{\ln(x)}{\sqrt{x}}$. 50 [50%]

(a) $$\lim_{x \to 0} \frac{e^{\sin(x)} - 1 - x}{x^2} = \lim_{x \to 0} \frac{\cos(x) e^{\sin(x)} - 1}{2x}$$

$$= \lim_{x \to 0} \frac{-\sin(x) e^{\sin(x)} + \cos^2(x) e^{\sin(x)}}{2} = \frac{1}{2}$$

(b) $$\lim_{x \to \infty} \frac{\ln x}{\sqrt{x}} = \lim_{x \to \infty} \frac{\frac{1}{x}}{\frac{1}{2x^{\frac{1}{2}}}} = \lim_{x \to \infty} \frac{2x^{\frac{1}{2}}}{x} = \lim_{x \to \infty} \frac{2}{x^{\frac{1}{2}}}$$

$$= 0$$